

GreenThumb

Reimagining Urban Community Gardens & Intelligent Plant Care

A human-centered mobile app & web ecosystem integrating smart hydration telemetry, neighborhood plot management, and local botanical exchange.

Assignment 4 Final

Interactive Prototype

WCAG AAA Design System



● Smart Telemetry

Real-time moisture & watering tracking

● Plot Management

Reserve & coordinate community beds

● Seed Marketplace

Local exchange for seeds & supplies

Why Urban Plant Care is Broken

Urban gardeners face three critical friction points that lead to plant neglect.

1. The Care & Watering Telemetry Gap

PROBLEM: Over 70% of houseplant deaths stem from erratic watering routines, guessing soil moisture, and improper sunlight exposure.

SOLUTION: Dynamic moisture telemetry gauges, watering reminders, and EasyCare diagnostic cards for proactive health management.

2. Fragmented Community Plot Coordination

PROBLEM: Neighborhood garden plots are manually booked on spreadsheets or bulletin boards with zero real-time availability.

SOLUTION: Interactive community garden directory (e.g., Plot 12 North Section) with automated waitlists and volunteer logs.

3. Scattered Resources & Supply Barriers

PROBLEM: Gardeners navigate disjointed stores for verified seed kits, organic compost, and expert pruning workshops.

SOLUTION: Integrated local nursery marketplace and masterclass calendar with RSVP ticketing directly inside the application.

Peer-to-Peer Marketplace & Events

Connecting urban growers with verified nurseries, seed swaps, and workshops.

● Local Nursery & Supply Marketplace

Direct-to-grower catalog for organic fertilizers, heirloom seeds, and starter sets.



● Community Masterclasses & Plot Reservations

Hands-on pruning workshops, seed swap schedules, and shared plot waitlists.



Design System & WCAG AAA Standards

A production-ready token library built for clarity, contrast, and consistency.

Botanical Color Tokens & Contrast Specs

Deep Emerald

#144D37 • Brand / Nav Header

Forest Leaf

#2C8B59 • Primary Action CTAs

Sage Mint

#85C19C • Status Pills & Badges

Terracotta

#E76F51 • Alerts & Harvest Cards

Golden Sand

#F4A261 • Sunlight & Loyalty Points

Dark Slate

#1B2E26 • High-Contrast Typography

Ergonomic Thumb-Zone

Navigation is focused in the lower third of the screen for effortless single-handed mobile interaction and fast access.

[Home] [Explore] [Care] [Plots] [Profile]

Key Usability Metrics:

- WCAG 2.1 AAA Contrast
- 48px Minimum Tap Targets
- Tactile Haptic Responses
- Dynamic Font Scaling (12-36px)

Reusable Component Library

Standardized states built in Figma including active, hover, disabled, and progress rings for telemetry.

Filled Primary Button

Active Status Chip: 96% Optimal

Alert Badge: Water in 2h

Sunlight Dial: Bright Indirect

Experience the Full Prototype

Interactive Figma Files & Open Source Repository

Project Deliverables & Artifacts

Phase 4 (Final Deliverable)

AppPrototype_UIUX_Sheikh_Assignment4.fig • Final Mobile Prototype & Component System

Phase 3 (Interactive Web)

Prototype_Assignment3_UIUX_SheikhNaim.fig • High-Fidelity Web Hub & Community Modules

Phase 2 (Architecture)

SheikhNaim_UI_UX_Assignment2.fig • Responsive Web Wireframes & Grid Specs

Phase 1 (Research & Brief)

SheikhNaim-Assignment1.docx • UX Research Document & Whiteboard Low-Fi

View Source & Interactive Prototype

GitHub Repository:

github.com/snaimio/greenthumb-uiux-prototype

Figma Interactive Share URL:

<https://www.figma.com/design/7nuclqgy1E88gaqP8KAQKA/...>

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